Solution to the Daily Limit

Based on the problem provided in the file dailyintegral-limit-limit-1x1-2026-08-17.jpg, we are tasked with evaluating the following limit:

$$L = \lim_{n\to\infty} \sum_{k=1}^{n} \sum_{j=1}^{n} \frac{2k^{-\frac{2j}{n}} \ln(k)}{nk^2}$$

1. Evaluating the Inner Sum

First, let us isolate the inner summation which is over the index j. Since the terms involving n and k are independent of j, we can factor them out:

$$\sum_{j=1}^{n} \frac{2k^{-\frac{2j}{n}} \ln(k)}{nk^2} = \frac{2\ln(k)}{nk^2} \sum_{j=1}^{n} \left(k^{-\frac{2}{n}}\right)^j$$

The summation on the right is a standard geometric series with the common ratio $r = k^{-\frac{2}{n}}$. Note that for $k = 1$, the entire expression evaluates to zero because $\ln(1) = 0$. Thus, we only need to evaluate this for $k \geq 2$. Using the geometric sum formula $\sum_{j=1}^{n} r^j = \frac{r - r^{n+1}}{1-r} = \frac{1-r^n}{r^{-1}-1}$, we get:

$$\sum_{j=1}^{n} \left(k^{-\frac{2}{n}}\right)^j = \frac{1 - \left(k^{-\frac{2}{n}}\right)^n}{k^{\frac{2}{n}} - 1} = \frac{1 - k^{-2}}{k^{\frac{2}{n}} - 1}$$

Substitute this back into the expression for L:

$$L = \lim_{n\to\infty} \sum_{k=1}^{n} \frac{2\ln(k)}{nk^2} \left(\frac{1 - k^{-2}}{k^{\frac{2}{n}} - 1}\right)$$

By distributing the k^2 in the denominator and rearranging, we can rewrite the terms as:

$$L = \lim_{n\to\infty} \sum_{k=1}^{n} \left(\frac{1}{k^2} - \frac{1}{k^4}\right) \frac{\frac{2\ln(k)}{n}}{k^{\frac{2}{n}} - 1}$$

2. Applying the Limit

To evaluate the limit as $n \to \infty$, we observe the behavior of the fractional term involving n. We can rewrite $k^{\frac{2}{n}}$ using the exponential function:

$$k^{\frac{2}{n}} = e^{\frac{2\ln(k)}{n}}$$

Let $x = \frac{2\ln(k)}{n}$. The fraction becomes $\frac{x}{e^x - 1}$. We know that for $x > 0$, the function $f(x) = \frac{x}{e^x-1}$ is strictly bounded between 0 and 1. Consequently, each term in our sum is bounded by a convergent sequence:

$$\left| \left(\frac{1}{k^2} - \frac{1}{k^4}\right) \frac{\frac{2\ln(k)}{n}}{e^{\frac{2\ln(k)}{n}} - 1} \right| \leq \frac{1}{k^2}$$

Because $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges, Tannery's Theorem (or the Dominated Convergence Theorem for counting measures) allows us to interchange the limit and the summation:

$$L = \sum_{k=1}^{\infty} \lim_{n\to\infty} \left[\left(\frac{1}{k^2} - \frac{1}{k^4} \right) \frac{\frac{2\ln(k)}{n}}{e^{\frac{2\ln(k)}{n}} - 1} \right]$$

Using the fundamental limit $\lim_{x\to 0} \frac{x}{e^x - 1} = 1$, we evaluate the inner limit:

$$\lim_{n\to\infty} \frac{\frac{2\ln(k)}{n}}{e^{\frac{2\ln(k)}{n}} - 1} = 1$$

Conclusion

Substituting this result back into the sum simplifies the expression dramatically:

$$L = \sum_{k=1}^{\infty} \left(\frac{1}{k^2} - \frac{1}{k^4} \right) = \sum_{k=1}^{\infty} \frac{1}{k^2} - \sum_{k=1}^{\infty} \frac{1}{k^4}$$

These are well-known values of the Riemann zeta function, $\zeta(s) = \sum_{k=1}^{\infty} \frac{1}{k^s}$, evaluated at $s = 2$ and $s = 4$:

$$\zeta(2) = \frac{\pi^2}{6} \quad \text{and} \quad \zeta(4) = \frac{\pi^4}{90}$$

Therefore, the exact value of the limit is:

$$\lim_{n\to\infty} \sum_{k=1}^{n} \sum_{j=1}^{n} \frac{2k^{-\frac{2j}{n}} \ln(k)}{nk^2} = \frac{\pi^2}{6} - \frac{\pi^4}{90}$$